

M21 2026 July 2026 N250 Component

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1 Setup

Load libraries

Set emmeans options

```
emm_options(  
  lmer.df = "kenward-roger",  
  pbkrtest.limit = 10000,  
  disable.pbkrtest = FALSE  
)
```

Set ggplot parameters

```
# Color palette  
my_palette_1 <- c(  
  "#E31A1C", # red  
  "#1F78B4" # blue  
)  
  
my_palette_2 <- c(  
  "#0072B2", # dark blue  
  "#E69F00" # orange/gold  
)  
my_linetypes_1 <- c("dashed", "solid")  
  
# Theme  
my_theme <- theme_classic(base_size = 10) +  
  theme(  
    legend.position = "bottom",  
    legend.direction = "horizontal",  
    legend.box = "horizontal"  
  )  
  
# Common guides  
my_guides <- guides(  
  colour = guide_legend(nrow = 1),  
  fill = guide_legend(nrow = 1)  
)
```

Define 2-way interaction table function

2 Load Files

```
# Load data: rt words
```

```

rt_w <- read_csv("rt_words_frq.csv") |>
  filter(correct == 1) |>
  mutate(
    SubjID = factor(SubjID),
    Item = factor(target),
    logRT = log(response_time),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1),
    zLogBF = as.numeric(scale(LogBF)),
    zLogFS = as.numeric(scale(LogFS))
  )

# Load data: rt nonwords
rt_nw <- read_csv("rt_nwords_frq.csv") |>
  filter(correct == 1) |>
  mutate(
    SubjID = factor(SubjID),
    ItemID = factor(ItemID),
    complexity = factor(complexity,
      levels = c("SIMP", "COMP"),
      labels = c("Simple", "Complex")),
    logRT = log(response_time),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1),
    zLogBF = as.numeric(scale(LogBF)),
    zLogFS = as.numeric(scale(LogFS))
  )

# Load data: erp_fs words
erp_fs_w <- read_csv("erp_fs_dmg_words.csv") |>
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    family_size = factor(family_size,
      levels = c("Small", "Large")),
    time_window = factor(time_window),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )

# Load data: erp_fs nonwords
erp_fs_nw <- read_csv("erp_fs_dmg_nonwords.csv") |>
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    family_size = factor(family_size,
      levels = c("Small", "Large")),
    complexity = factor(complexity,
      labels = c("Simple", "Complex")),
    time_window = factor(time_window),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )

# Load data: erp_bf words
erp_bf_w <- read_csv("erp_bf_dmg_words.csv") |>
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    base_frequency = factor(base_frequency,
      levels = c("Low", "High")),
    time_window = factor(time_window),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )

# Load data: erp_bf nonwords
erp_bf_nw <- read_csv("erp_bf_dmg_nonwords.csv") |>
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    base_frequency = factor(base_frequency,
      levels = c("Low", "High")),
    complexity = factor(complexity,
      labels = c("Simple", "Complex")),
    time_window = factor(time_window),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )

```

3 N250 WORD DATA: INTERACTION WITH OS (PRIMARY ANALYSES)

3.1 N250 WORD DATA: FAMILY SIZE

3.1.1 Model

```

contrasts(erp_fs_w$family_size) <- c(-0.5, 0.5)

m_w_N250_fs <- lmer(
  value ~ family_size * OS + family_size * LP +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_fs_w, time_window == "N250"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_w_N250_fs)

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ family_size * OS + family_size * LP + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_fs_w, time_window == "N250")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 15028.7
||
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -8.4814 -0.4012 -0.0214  0.3759 12.9242
||
|| Random effects:
|| Groups      Name                Variance Std.Dev.
|| SubjID:chlabel (Intercept)  4.626      2.151
|| SubjID       (Intercept)  2.918      1.708
|| Residual                    2.752      1.659
|| Number of obs: 3240, groups: SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||              Estimate Std. Error      df t value Pr(>|t|)
|| (Intercept)   -0.75979    0.22921   57.00012   -3.315   0.0016 **
|| family_size1    0.32016    0.05839 1616.99995    5.483 4.84e-08 ***
|| OS              0.26085    0.25635   57.00012    1.018   0.3132
|| LP             -0.27860    0.18636   57.00012   -1.495   0.1404
|| family_size:OS  0.10911    0.06531 1616.99995    1.671   0.0950 .
|| family_size:LP -0.07246    0.04748 1616.99995   -1.526   0.1272
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||      (Intr) fmly_1 OS      LP      f_1:OS
|| family_size1  0.000
|| OS           -0.046  0.000
|| LP            0.038  0.000  0.021
|| fmly_sz1:OS  0.000 -0.046  0.000  0.000
|| fmly_sz1:LP  0.000  0.038  0.000  0.000  0.021
||
|| # Effect sizes for individual terms
eta_squared(m_w_N250_fs, partial = TRUE)

|| # Effect Size for ANOVA (Type III)
||
|| Parameter      | Eta2 (partial) | 95% CI
|| -----
|| family_size    | 0.02 | [0.01, 1.00]
|| OS              | 0.02 | [0.00, 1.00]
|| LP              | 0.04 | [0.00, 1.00]
|| family_size:OS | 1.72e-03 | [0.00, 1.00]
|| family_size:LP | 1.44e-03 | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].
||
|| # Model R²
r2_nakagawa(m_w_N250_fs)

|| # R2 for Mixed Models
||
|| Conditional R2: 0.738
|| Marginal R2: 0.020
Model Comparison

```

```

contrasts(erp_fs_w$family_size) <- c(-0.5, 0.5)

# Model with the two individual-difference interactions with family size
m_base_fs <- lmer(
  value ~ family_size * OS +
    family_size * LP +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_fs_w, time_window == "N250"),
  REML = FALSE,
  control = lmerControl(optimizer = "bobyqa")
)

# Add OS x LP
m_2way_fs <- update(
  m_base_fs,
  . ~ . + OS:LP
)

# Add the three-way interaction
m_3way_fs <- update(
  m_2way_fs,
  . ~ . + family_size:OS:LP
)

anova(m_base_fs, m_2way_fs, m_3way_fs)

|| Data: subset(erp_fs_w, time_window == "N250")
|| Models:
|| m_base_fs: value ~ family_size * OS + family_size * LP + (1 | SubjID) + (1 | SubjID:chlabel)
|| m_2way_fs: value ~ family_size * OS + LP + (1 | SubjID) + (1 | SubjID:chlabel) + family_size:OS + family_size:LP + OS:LP
|| m_3way_fs: value ~ family_size * OS + LP + (1 | SubjID) + (1 | SubjID:chlabel) + family_size:OS + family_size:LP + OS:LP + family_size:OS:LP
||      npar   AIC    BIC logLik deviance   Chisq Df Pr(>Chisq)
|| m_base_fs    9 15031 15086 -7506.7    15013
|| m_2way_fs   10 15033 15094 -7506.4    15013  0.6598  1    0.4166
|| m_3way_fs   11 15015 15082 -7496.5   14993 19.8565  1  8.348e-06 ***
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

summary(m_3way_fs)

|| Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ family_size + OS + LP + (1 | SubjID) + (1 | SubjID:chlabel) + family_size:OS + family_size:LP + OS:LP + family_size:OS:LP
|| Data: subset(erp_fs_w, time_window == "N250")
|| Control: lmerControl(optimizer = "bobyqa")
||
||      AIC      BIC logLik deviance df.resid
|| 15014.9 15081.8 -7496.5 14992.9    3229
||
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -8.5376 -0.4090 -0.0270  0.3679 12.9025
||
|| Random effects:
|| Groups      Name                Variance Std.Dev.
|| SubjID:chlabel (Intercept)  4.646     2.155
|| SubjID        (Intercept)  2.729     1.652
|| Residual                        2.713     1.647
|| Number of obs: 3240, groups: SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||              Estimate Std. Error    df t value Pr(>|t|)
|| (Intercept)   -0.76558    0.22229  60.00000  -3.444  0.00105 **
|| family_size1    0.31188    0.05801 1620.00000   5.376 8.71e-08 ***
|| OS              0.26269    0.24850  60.00000   1.057  0.29471
|| LP             -0.29475    0.18173  60.00000  -1.622  0.11006
|| family_size1:OS  0.11173    0.06485 1620.00000   1.723  0.08509 .
|| family_size1:LP -0.09558    0.04743 1620.00000  -2.015  0.04402 *
|| OS:LP          -0.19686    0.24170  60.00000  -0.815  0.41858
|| family_size1:OS:LP -0.28193    0.06308 1620.00000  -4.470 8.37e-06 ***
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||      (Intr) fmly_1 OS      LP      fm_1:OS f_1:LP OS:LP
|| family_size1  0.000
|| OS           -0.046  0.000
|| LP            0.041  0.000  0.020
|| fmly_sz1:OS  0.000 -0.046  0.000  0.000

```

```

|| fml_y_sz1:LP 0.000 0.041 0.000 0.000 0.020
|| OS:LP 0.032 0.000 -0.009 0.109 0.000 0.000
|| fml_1:OS:LP 0.000 0.032 0.000 0.000 -0.009 0.109 0.000
library(emmeans)
emm_options(lmer.df = "satterthwaite")

# Family Size three-way follow-up: FS simple effect at Low/Mean/High OS x Low/Mean/High LP
# Representative values: mean and ±1 SD for each continuous IV
os_vals_fs <- c(mean(erp_fs_w$OS) - sd(erp_fs_w$OS), mean(erp_fs_w$OS), mean(erp_fs_w$OS) + sd(erp_fs_w$OS))
lp_vals_fs <- c(mean(erp_fs_w$LP) - sd(erp_fs_w$LP), mean(erp_fs_w$LP), mean(erp_fs_w$LP) + sd(erp_fs_w$LP))

emm_fs <- emmeans(m_3way_fs, ~ family_size | OS * LP,
  at = list(OS = os_vals_fs, LP = lp_vals_fs))
fs_followup <- contrast(emm_fs, method = "revpairwise", by = c("OS", "LP"))
summary(fs_followup, infer = c(TRUE, TRUE))

|| OS = -0.8509, LP = -1.2763:
|| contrast estimate SE df lower.CL upper.CL t.ratio p.value
|| Large - Small 0.03260 0.1200 1624 -0.2024 0.268 0.272 0.7856
||
|| OS = 0.0418, LP = -1.2763:
|| contrast estimate SE df lower.CL upper.CL t.ratio p.value
|| Large - Small 0.45358 0.0824 1624 0.2920 0.615 5.507 <.0001
||
|| OS = 0.9345, LP = -1.2763:
|| contrast estimate SE df lower.CL upper.CL t.ratio p.value
|| Large - Small 0.87456 0.1280 1624 0.6231 1.126 6.821 <.0001
||
|| OS = -0.8509, LP = -0.0483:
|| contrast estimate SE df lower.CL upper.CL t.ratio p.value
|| Large - Small 0.20984 0.0821 1624 0.0488 0.371 2.556 0.0107
||
|| OS = 0.0418, LP = -0.0483:
|| contrast estimate SE df lower.CL upper.CL t.ratio p.value
|| Large - Small 0.32173 0.0580 1624 0.2080 0.435 5.550 <.0001
||
|| OS = 0.9345, LP = -0.0483:
|| contrast estimate SE df lower.CL upper.CL t.ratio p.value
|| Large - Small 0.43363 0.0820 1624 0.2728 0.594 5.289 <.0001
||
|| OS = -0.8509, LP = 1.1798:
|| contrast estimate SE df lower.CL upper.CL t.ratio p.value
|| Large - Small 0.38707 0.1140 1624 0.1638 0.610 3.401 0.0007
||
|| OS = 0.0418, LP = 1.1798:
|| contrast estimate SE df lower.CL upper.CL t.ratio p.value
|| Large - Small 0.18988 0.0827 1624 0.0277 0.352 2.296 0.0218
||
|| OS = 0.9345, LP = 1.1798:
|| contrast estimate SE df lower.CL upper.CL t.ratio p.value
|| Large - Small -0.00731 0.1270 1624 -0.2566 0.242 -0.058 0.9541
||
|| Degrees-of-freedom method: kenward-roger
|| Confidence level used: 0.95

# family size
as.tibble(emmeans(m_w_N250_fs, ~ family_size)) |> print()

```

Means and Contrasts

```

|| # A tibble: 2 x 6
|| family_size emmean SE df lower.CL upper.CL
|| <fct> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 Small -0.900 0.231 58.9 -1.36 -0.438
|| 2 Large -0.571 0.231 58.9 -1.03 -0.110
emm_fs_OS <- emmeans(m_w_N250_fs,
  ~ family_size | OS,
  at = list(OS = c(-1.9, -1, 0, 1, 1.9)),
  lmerTest.limit = 6480)

emm_part <- summary(emm_fs_OS) |>
  as.data.frame() |>
  transmute(OS, Effect = paste("EMM:", family_size), Estimate = emmean, SE, df,
    lower.CL, upper.CL, t.ratio = NA_real_, p.value = NA_real_)

contrast_part <- pairs(emm_fs_OS) |>
  summary() |>
  as.data.frame() |>

```

```

transmute(OS, Effect = paste("Contrast:", contrast), Estimate = estimate, SE, df,
          lower.CL = NA_real_, upper.CL = NA_real_, t.ratio, p.value)

emm_contrast_table <- bind_rows(emm_part, contrast_part) |>
  arrange(OS, Effect)|>
  mutate(
    Estimate = round(Estimate, 2),
    SE       = round(SE, 2),
    df       = round(df, 1),
    lower.CL = round(lower.CL, 2),
    upper.CL = round(upper.CL, 2),
    t.ratio  = round(t.ratio, 2),
    p.value  = round(p.value, 3)
  )

as.tibble(emm_contrast_table, digits = 3) |> print(n = 20)

|| # A tibble: 15 x 9
||   OS Effect      Estimate SE    df lower.CL upper.CL t.ratio p.value
||   <dbl> <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 -1.9 Contrast: Small - Large -0.12 0.14 1617 NA NA -0.83 0.405
|| 2 -1.9 EMM: Large -1.18 0.55 58.9 -2.29 -0.08 NA NA
|| 3 -1.9 EMM: Small -1.3 0.55 58.9 -2.41 -0.2 NA NA
|| 4 -1 Contrast: Small - Large -0.21 0.09 1617 NA NA -2.39 0.017
|| 5 -1 EMM: Large -0.9 0.35 58.9 -1.61 -0.19 NA NA
|| 6 -1 EMM: Small -1.11 0.35 58.9 -1.82 -0.41 NA NA
|| 7 0 Contrast: Small - Large -0.32 0.06 1617 NA NA -5.55 0
|| 8 0 EMM: Large -0.58 0.23 58.9 -1.05 -0.12 NA NA
|| 9 0 EMM: Small -0.91 0.23 58.9 -1.37 -0.45 NA NA
|| 10 1 Contrast: Small - Large -0.43 0.09 1617 NA NA -5.06 0
|| 11 1 EMM: Large -0.27 0.34 58.9 -0.95 0.41 NA NA
|| 12 1 EMM: Small -0.7 0.34 58.9 -1.38 -0.02 NA NA
|| 13 1.9 Contrast: Small - Large -0.53 0.13 1617 NA NA -3.94 0
|| 14 1.9 EMM: Large 0.01 0.53 58.9 -1.05 1.08 NA NA
|| 15 1.9 EMM: Small -0.52 0.53 58.9 -1.58 0.55 NA NA

# family_size
as.tibble(emmeans(m_w_N250_fs, ~ family_size)) |> print()

|| # A tibble: 2 x 6
||   family_size emmean SE    df lower.CL upper.CL
||   <fct>      <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 Small -0.900 0.231 58.9 -1.36 -0.438
|| 2 Large -0.571 0.231 58.9 -1.03 -0.110

emm_fs_LP <- emmeans(m_w_N250_fs,
  ~ family_size | LP,
  at = list(LP = c(-2.5, -1, 0, 1, 3.0)),
  lmerTest.limit = 6480)

emm_part <- summary(emm_fs_LP) |>
  as.data.frame() |>
  transmute(LP, Effect = paste("EMM:", family_size), Estimate = emmean, SE, df,
            lower.CL, upper.CL, t.ratio = NA_real_, p.value = NA_real_)

contrast_part <- pairs(emm_fs_LP) |>
  summary() |>
  as.data.frame() |>
  transmute(LP, Effect = paste("Contrast:", contrast), Estimate = estimate, SE, df,
            lower.CL = NA_real_, upper.CL = NA_real_, t.ratio, p.value)

emm_contrast_table <- bind_rows(emm_part, contrast_part) |>
  arrange(LP, Effect)|>
  mutate(
    Estimate = round(Estimate, 2),
    SE       = round(SE, 2),
    df       = round(df, 1),
    lower.CL = round(lower.CL, 2),
    upper.CL = round(upper.CL, 2),
    t.ratio  = round(t.ratio, 2),
    p.value  = round(p.value, 3)
  )

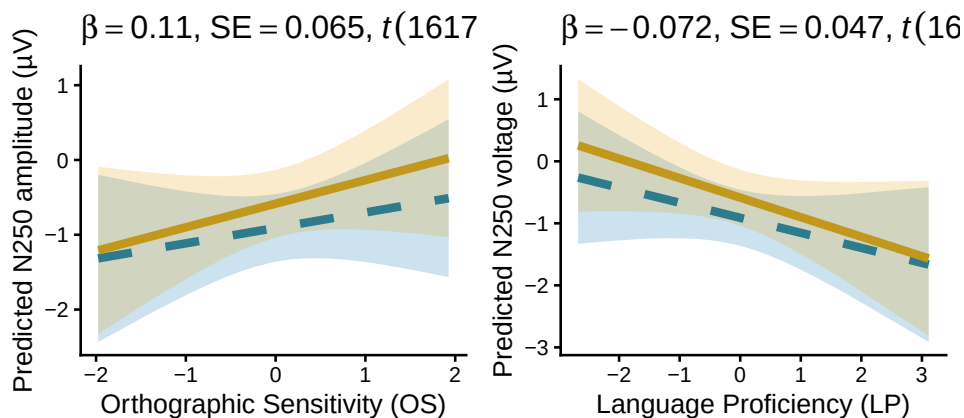
as.tibble(emm_contrast_table, digits = 3) |> print(n = 20)

|| # A tibble: 15 x 9
||   LP Effect      Estimate SE    df lower.CL upper.CL t.ratio p.value
||   <dbl> <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 -2.5 Contrast: Small - Large -0.51 0.13 1617 NA NA -3.89 0
|| 2 -2.5 EMM: Large 0.2 0.52 58.9 -0.83 1.23 NA NA
|| 3 -2.5 EMM: Small -0.31 0.52 58.9 -1.34 0.73 NA NA

```

4	-1	Contrast: Small - Large	-0.4	0.07	1617	NA	NA	-5.39	0
5	-1	EMM: Large	-0.27	0.29	58.9	-0.86	0.31	NA	NA
6	-1	EMM: Small	-0.67	0.29	58.9	-1.25	-0.08	NA	NA
7	0	Contrast: Small - Large	-0.32	0.06	1617	NA	NA	-5.57	0
8	0	EMM: Large	-0.59	0.23	58.9	-1.05	-0.12	NA	NA
9	0	EMM: Small	-0.91	0.23	58.9	-1.37	-0.45	NA	NA
10	1	Contrast: Small - Large	-0.25	0.08	1617	NA	NA	-3.29	0.001
11	1	EMM: Large	-0.9	0.3	58.9	-1.51	-0.29	NA	NA
12	1	EMM: Small	-1.15	0.3	58.9	-1.76	-0.55	NA	NA
13	3	Contrast: Small - Large	-0.11	0.16	1617	NA	NA	-0.69	0.492
14	3	EMM: Large	-1.53	0.62	58.9	-2.77	-0.3	NA	NA
15	3	EMM: Small	-1.64	0.62	58.9	-2.87	-0.4	NA	NA

N250 Word Data



Plots Family Size  Small  Large Family Size  Small  Large ## N250 WORD
DATA: BASE FREQUENCY as Fixed Effect

3.1.2 Model

```
contrasts(erp_bf_w$base_frequency) <- c(-0.5, 0.5)

m_w_N250_bf <- lmer(
  value ~ base_frequency * OS + base_frequency * LP +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_bf_w, time_window == "N250"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_w_N250_bf)

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ base_frequency * OS + base_frequency * LP + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_bf_w, time_window == "N250")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 18997.8
||
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -7.3869 -0.3872 -0.0036  0.3943 14.0848
||
|| Random effects:
|| Groups      Name                Variance Std.Dev.
|| SubjID:chlabel (Intercept) 19.732   4.442
|| SubjID       (Intercept) 12.044   3.470
|| Residual                    8.078   2.842
|| Number of obs: 3240, groups: SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||              Estimate Std. Error      df t value Pr(>|t|)
|| (Intercept)   -1.50586    0.46497   57.00001  -3.239  0.00201 **
|| base_frequency1 -0.31021    0.10004 1617.00007  -3.101  0.00196 **
|| OS              0.53410    0.52004   57.00001   1.027  0.30874
|| LP             -0.53556    0.37805   57.00000  -1.417  0.16203
|| base_frequency1:OS -0.44996    0.11189 1617.00007  -4.021 6.05e-05 ***
```

```

|| base_frequency1:LP    0.17285    0.08134 1617.00007    2.125  0.03374 *
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||      (Intr) bs_fr1 OS      LP      b_1:OS
|| bas_frqn1  0.000
|| OS        -0.046  0.000
|| LP         0.038  0.000  0.021
|| bs_frqn1:OS 0.000 -0.046  0.000  0.000
|| bs_frqn1:LP 0.000  0.038  0.000  0.000  0.021

```

Effect sizes for individual terms
eta_squared(m_w_N250_bf, partial = TRUE)

```

|| # Effect Size for ANOVA (Type III)
||
|| Parameter          | Eta2 (partial) |      95% CI
|| -----
|| base_frequency    |      5.91e-03 | [0.00, 1.00]
|| OS                |           0.02 | [0.00, 1.00]
|| LP                |           0.03 | [0.00, 1.00]
|| base_frequency:OS |      9.90e-03 | [0.00, 1.00]
|| base_frequency:LP |      2.78e-03 | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].

```

Model R²
r2_nakagawa(m_w_N250_bf)

```

|| # R2 for Mixed Models
||
|| Conditional R2: 0.801
|| Marginal R2: 0.019

```

Model Comparison
contrasts(erp_bf_w\$base_frequency) <- c(-0.5, 0.5)

Model with the two individual-difference interactions with base frequency
m_base_bf <- lmer(
 value ~ base_frequency * OS +
 base_frequency * LP +
 (1 | SubjID) +
 (1 | SubjID:chlabel),
 data = subset(erp_bf_w, time_window == "N250"),
 REML = FALSE,
 control = lmerControl(optimizer = "bobyqa")
)

Add OS × LP
m_2way_bf <- update(
 m_base_bf,
 . ~ . + OS:LP
)

Add the three-way interaction
m_3way_bf <- update(
 m_2way_bf,
 . ~ . + base_frequency:OS:LP
)

anova(m_base_bf, m_2way_bf, m_3way_bf)

```

|| Data: subset(erp_bf_w, time_window == "N250")
|| Models:
|| m_base_bf: value ~ base_frequency * OS + base_frequency * LP + (1 | SubjID) + (1 | SubjID:chlabel)
|| m_2way_bf: value ~ base_frequency + OS + LP + (1 | SubjID) + (1 | SubjID:chlabel) + base_frequency:OS + base_frequency:LP + OS:LP
|| m_3way_bf: value ~ base_frequency + OS + LP + (1 | SubjID) + (1 | SubjID:chlabel) + base_frequency:OS + base_frequency:LP + OS:LP + base_frequency:OS:LP
||      npar  AIC    BIC  logLik deviance   Chisq Df Pr(>Chisq)
|| m_base_bf    9 19008 19063 -9495.0   18990
|| m_2way_bf   10 19009 19070 -9494.7   18989  0.6082  1    0.4355
|| m_3way_bf   11 18987 19054 -9482.4   18965 24.5465  1  7.254e-07 ***
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

summary(m_3way_bf)

```

|| Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ base_frequency + OS + LP + (1 | SubjID) + (1 | SubjID:chlabel) +
||      base_frequency:OS + base_frequency:LP + OS:LP + base_frequency:OS:LP
|| Data: subset(erp_bf_w, time_window == "N250")

```

```

|| Control: lmerControl(optimizer = "bobyqa")
||
||      AIC      BIC   logLik deviance df.resid
|| 18986.8 19053.7 -9482.4 18964.8    3229
||
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -7.3174 -0.3952 -0.0067  0.4000 14.0874
||
|| Random effects:
||  Groups      Name      Variance Std.Dev.
||  SubjID:chlabel (Intercept) 19.800  4.450
||  SubjID        (Intercept) 11.274  3.358
||  Residual                        7.941  2.818
|| Number of obs: 3240, groups: SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||              Estimate Std. Error      df t value Pr(>|t|)
|| (Intercept)    -1.51713    0.45113   60.00009   -3.363  0.00135 **
|| base_frequency1 -0.32598    0.09925 1619.99975   -3.285  0.00104 **
|| OS              0.53767    0.50433   60.00009    1.066  0.29064
|| LP             -0.56702    0.36882   60.00009   -1.537  0.12945
|| base_frequency1:OS -0.44496    0.11095 1619.99975   -4.010 6.34e-05 ***
|| base_frequency1:LP  0.12883    0.08114 1619.99975    1.588  0.11252
|| OS:LP           -0.38351    0.49052   60.00009   -0.782  0.43739
|| base_frequency1:OS:LP -0.53668    0.10791 1619.99975   -4.973 7.28e-07 ***
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||      (Intr) bs_fr1 OS      LP      bs_1:OS b_1:LP OS:LP
|| bas_frqn1 0.000
|| OS        -0.046 0.000
|| LP         0.041 0.000 0.020
|| bs_frqn1:OS 0.000 -0.046 0.000 0.000
|| bs_frqn1:LP 0.000 0.041 0.000 0.000 0.020
|| OS:LP       0.032 0.000 -0.009 0.109 0.000 0.000
|| bs_f1:OS:LP 0.000 0.032 0.000 0.000 -0.009 0.109 0.000

```

```

library(emmeans)
emm_options(lmer.df = "satterthwaite")

# Base Frequency three-way follow-up
# Representative values: mean and ±1 SD for each continuous IV
os_vals_bf <- c(mean(erp_bf_w$OS) - sd(erp_bf_w$OS), mean(erp_bf_w$OS), mean(erp_bf_w$OS) + sd(erp_bf_w$OS))
lp_vals_bf <- c(mean(erp_bf_w$LP) - sd(erp_bf_w$LP), mean(erp_bf_w$LP), mean(erp_bf_w$LP) + sd(erp_bf_w$LP))

emm_bf <- emmeans(m_3way_bf, ~ base_frequency | OS * LP,
  at = list(OS = os_vals_bf, LP = lp_vals_bf))
bf_followup <- contrast(emm_bf, method = "revpairwise", by = c("OS", "LP"))
summary(bf_followup, infer = c(TRUE, TRUE))

```

```

|| OS = -0.8509, LP = -1.2763:
|| contrast estimate SE df lower.CL upper.CL t.ratio p.value
|| High - Low -0.6946 0.2050 1624 -1.097 -0.2926 -3.389 0.0007
||
|| OS = 0.0418, LP = -1.2763:
|| contrast estimate SE df lower.CL upper.CL t.ratio p.value
|| High - Low -0.4804 0.1410 1624 -0.757 -0.2040 -3.409 0.0007
||
|| OS = 0.9345, LP = -1.2763:
|| contrast estimate SE df lower.CL upper.CL t.ratio p.value
|| High - Low -0.2661 0.2190 1624 -0.696 0.1641 -1.213 0.2252
||
|| OS = -0.8509, LP = -0.0483:
|| contrast estimate SE df lower.CL upper.CL t.ratio p.value
|| High - Low 0.0244 0.1400 1624 -0.251 0.2999 0.174 0.8621
||
|| OS = 0.0418, LP = -0.0483:
|| contrast estimate SE df lower.CL upper.CL t.ratio p.value
|| High - Low -0.3497 0.0992 1624 -0.544 -0.1552 -3.526 0.0004
||
|| OS = 0.9345, LP = -0.0483:
|| contrast estimate SE df lower.CL upper.CL t.ratio p.value
|| High - Low -0.7238 0.1400 1624 -0.999 -0.4487 -5.161 <.0001
||
|| OS = -0.8509, LP = 1.1798:
|| contrast estimate SE df lower.CL upper.CL t.ratio p.value
|| High - Low 0.7434 0.1950 1624 0.361 1.1254 3.818 0.0001
||
|| OS = 0.0418, LP = 1.1798:

```

```

|| contrast estimate SE df lower.CL upper.CL t.ratio p.value
|| High - Low -0.2190 0.1410 1624 -0.497 0.0585 -1.548 0.1218
||
|| OS = 0.9345, LP = 1.1798:
|| contrast estimate SE df lower.CL upper.CL t.ratio p.value
|| High - Low -1.1815 0.2170 1624 -1.608 -0.7550 -5.434 <.0001
||
|| Degrees-of-freedom method: kenward-roger
|| Confidence level used: 0.95

```

```
as.tibble(emmeans(m_w_N250_bf, ~ base_frequency)) |> print()
```

Means and contrasts

```

|| # A tibble: 2 x 6
|| base_frequency emmean SE df lower.CL upper.CL
|| <fct> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 Low -1.29 0.467 58.3 -2.22 -0.355
|| 2 High -1.63 0.467 58.3 -2.56 -0.692

emm_bf_OS <- emmeans(m_w_N250_bf,
  ~ base_frequency | OS,
  at = list(OS = c(-1.9, -1, 0, 1, 1.9)),
  lmerTest.limit = 6480)

emm_part <- summary(emm_bf_OS) |>
  as.data.frame() |>
  transmute(OS, Effect = paste("EMM:", base_frequency), Estimate = emmean, SE, df,
    lower.CL, upper.CL, t.ratio = NA_real_, p.value = NA_real_)

contrast_part <- pairs(emm_bf_OS) |>
  summary() |>
  as.data.frame() |>
  transmute(OS, Effect = paste("Contrast:", contrast), Estimate = estimate, SE, df,
    lower.CL = NA_real_, upper.CL = NA_real_, t.ratio, p.value)

emm_contrast_table <- bind_rows(emm_part, contrast_part) |>
  arrange(OS, Effect)|>
  mutate(
    Estimate = round(Estimate, 2),
    SE = round(SE, 2),
    df = round(df, 1),
    lower.CL = round(lower.CL, 2),
    upper.CL = round(upper.CL, 2),
    t.ratio = round(t.ratio, 2),
    p.value = round(p.value, 3)
  )

as.tibble(emm_contrast_table, digits = 3) |> print(n = 20)

```

```

|| # A tibble: 15 x 9
|| OS Effect Estimate SE df lower.CL upper.CL t.ratio p.value
|| <dbl> <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 -1.9 Contrast: Low - High -0.54 0.24 1617 NA NA -2.24 0.025
|| 2 -1.9 EMM: High -2.23 1.12 58.3 -4.46 0.01 NA NA
|| 3 -1.9 EMM: Low -2.76 1.12 58.3 -5 -0.53 NA NA
|| 4 -1 Contrast: Low - High -0.13 0.15 1617 NA NA -0.86 0.392
|| 5 -1 EMM: High -1.95 0.72 58.3 -3.38 -0.51 NA NA
|| 6 -1 EMM: Low -2.08 0.72 58.3 -3.52 -0.64 NA NA
|| 7 0 Contrast: Low - High 0.32 0.1 1617 NA NA 3.19 0.001
|| 8 0 EMM: High -1.64 0.47 58.3 -2.57 -0.7 NA NA
|| 9 0 EMM: Low -1.32 0.47 58.3 -2.26 -0.39 NA NA
|| 10 1 Contrast: Low - High 0.77 0.15 1617 NA NA 5.25 0
|| 11 1 EMM: High -1.33 0.68 58.3 -2.7 0.04 NA NA
|| 12 1 EMM: Low -0.56 0.68 58.3 -1.93 0.81 NA NA
|| 13 1.9 Contrast: Low - High 1.17 0.23 1617 NA NA 5.09 0
|| 14 1.9 EMM: High -1.05 1.08 58.3 -3.21 1.11 NA NA
|| 15 1.9 EMM: Low 0.12 1.08 58.3 -2.04 2.28 NA NA

```

```
as.tibble(emmeans(m_w_N250_bf, ~ base_frequency)) |> print()
```

```

|| # A tibble: 2 x 6
|| base_frequency emmean SE df lower.CL upper.CL
|| <fct> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 Low -1.29 0.467 58.3 -2.22 -0.355
|| 2 High -1.63 0.467 58.3 -2.56 -0.692

emm_bf_LP <- emmeans(m_w_N250_bf,
  ~ base_frequency | LP,
  at = list(LP = c(-1.9, -1, 0, 1, 1.9)),
  lmerTest.limit = 6480)

```

```

emm_part <- summary(emm_bf_LP) |>
  as.data.frame() |>
  transmute(LP, Effect = paste("EMM:", base_frequency), Estimate = emmean, SE, df,
    lower.CL, upper.CL, t.ratio = NA_real_, p.value = NA_real_)

contrast_part <- pairs(emm_bf_LP) |>
  summary() |>
  as.data.frame() |>
  transmute(LP, Effect = paste("Contrast:", contrast), Estimate = estimate, SE, df,
    lower.CL = NA_real_, upper.CL = NA_real_, t.ratio, p.value)

emm_contrast_table <- bind_rows(emm_part, contrast_part) |>
  arrange(LP, Effect) |>
  mutate(
    Estimate = round(Estimate, 2),
    SE = round(SE, 2),
    df = round(df, 1),
    lower.CL = round(lower.CL, 2),
    upper.CL = round(upper.CL, 2),
    t.ratio = round(t.ratio, 2),
    p.value = round(p.value, 3)
  )

as.tibble(emm_contrast_table, digits = 3) |> print(n = 20)

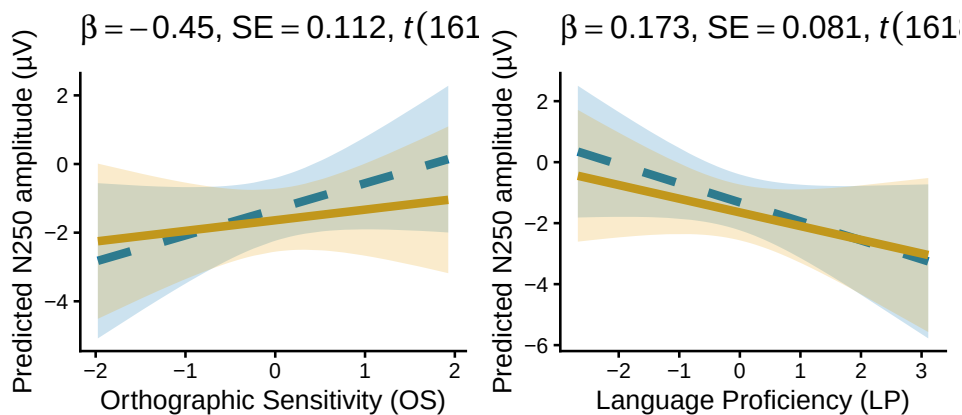
```

```

|| # A tibble: 15 x 9
||   LP Effect      Estimate    SE    df lower.CL upper.CL t.ratio p.value
||   <dbl> <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 -1.9 Contrast: Low - High    0.66  0.18 1617      NA      NA    3.64    0
|| 2 -1.9 EMM: High   -0.79  0.84  58.3   -2.49    0.9    NA     NA
|| 3 -1.9 EMM: Low   -0.14  0.84  58.3   -1.83    1.55    NA     NA
|| 4 -1 Contrast: Low - High    0.5   0.13 1617      NA      NA    3.97    0
|| 5 -1 EMM: High   -1.2   0.59  58.3   -2.38   -0.02    NA     NA
|| 6 -1 EMM: Low    -0.7   0.59  58.3   -1.88    0.49    NA     NA
|| 7 0 Contrast: Low - High    0.33  0.1 1617      NA      NA    3.29  0.001
|| 8 0 EMM: High   -1.65  0.47  58.3   -2.58   -0.71    NA     NA
|| 9 0 EMM: Low   -1.32  0.47  58.3   -2.25   -0.38    NA     NA
|| 10 0 Contrast: Low - High    0.16  0.13 1617      NA      NA    1.19  0.235
|| 11 1 EMM: High   -2.1   0.61  58.3   -3.33   -0.87    NA     NA
|| 12 1 EMM: Low   -1.94  0.61  58.3   -3.17   -0.71    NA     NA
|| 13 1.9 Contrast: Low - High    0     0.19 1617      NA      NA    0     0.997
|| 14 1.9 EMM: High  -2.5   0.88  58.3   -4.25   -0.75    NA     NA
|| 15 1.9 EMM: Low  -2.5   0.88  58.3   -4.25   -0.75    NA     NA

```

N250 Word Data



Plots Base Frequency Low High Base Frequency Low High